

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

THERMODYNAMICS

LECTURE-05 **PHYSICS**

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Topics To Be Covered

- 1** Revision
- 2** Gabbar Numerical Series PYQs
- 3** ..
- 4** ..
- 5** ..

Recap - ①

• Thermodynamics

- System
- Surrounding
- universe

✓ Intensive property

• Extensive property

✓ Isothermal

✓ Isobaric

✓ Isochoric

✓ Adiabatic

✓ Reversible

• Irreversible

✓ Cyclic process

• Temperature

✓ Thermal equilibrium

✓ Zeroth law of thermodynamics

Recap from 2nd Lecture

• Heat

$$Q_p = n C_p \Delta T \quad \overset{\text{Add}}{Q+} \quad Q-$$

• Internal Energy

$$\overset{Q_v}{U} = n C_v \Delta T \quad U+ \quad U-$$

• Work

Expansion $W = +ve$ Compression $W = -ive$

isobaric

• $W = P \Delta V$ ✓

Ideal gas equation

$$PV = nRT$$

$$P \Delta V = nR \Delta T$$

KTG



Recap from 3rd Lecture



✓ 1st law of thermodynamics ✓

$$Q = \Delta U + W$$

↓
law of conservation of Energy

Relationship b/w C_p and C_v

$$C_p - C_v = R$$

Heat Capacity Ratio (γ)

$$\gamma = \frac{C_p}{C_v}$$

WORK DONE BY Isothermal Process

$$W = 2.303 nRT \log \left(\frac{V_2}{V_1} \right)$$

Expansion Ratio.
lecture no. 4

④

WORK DONE BY Adiabatic Process

$$W = \frac{R}{(\gamma - 1)} (T_1 - T_2)$$

Heat engine

Carnot engine

$$\eta = 100\%$$

2nd law of thermodynamics



Doubts

Example:- **Internal Energy**

State Function :- the Quantity which only depend on **initial** and **Final point** ①
Do not depend on **path**.



$$\Delta U = U_f - U_i$$
$$= 300 - 100 = 200\text{ J}$$

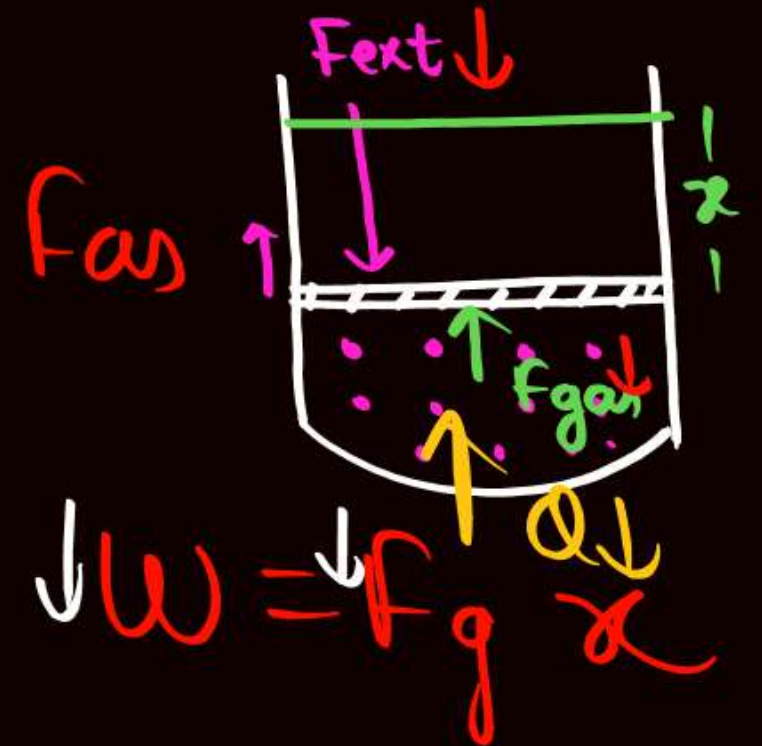
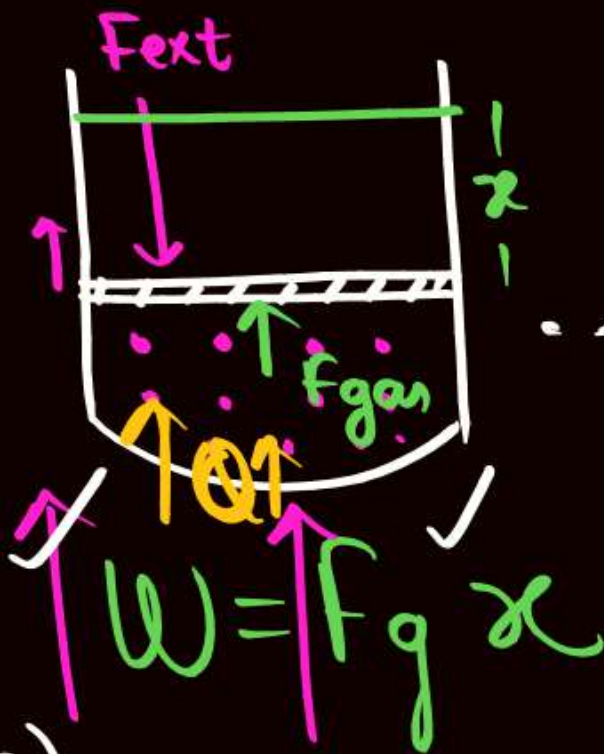
$V_i = 10\text{ lt}$ $V_f = 20\text{ lt}$
 $\Delta V = V_f - V_i$
 $= 20 - 10 = 10\text{ lt}$

$T_i = 25^\circ\text{C}$ $T_f = 35^\circ\text{C}$
 $\Delta T = T_f - T_i$
 $= 35 - 25$
 $= 10^\circ\text{C}$

Path Function :- the physical Quantity which depend on **path** (2)

Example :- WORK / Heat.

WORK ✓



Jammu

Srinagar

Banihal $x \downarrow$

Peer gali

Banihal — 180 Km (Distance)

Peer ki gali — 280 Km (Distance)

1st law of thermodynamic

③

$$\Rightarrow \text{Heat} = W + U$$

PF PF
Path Path
Function function

Doubt - (II)

(4)

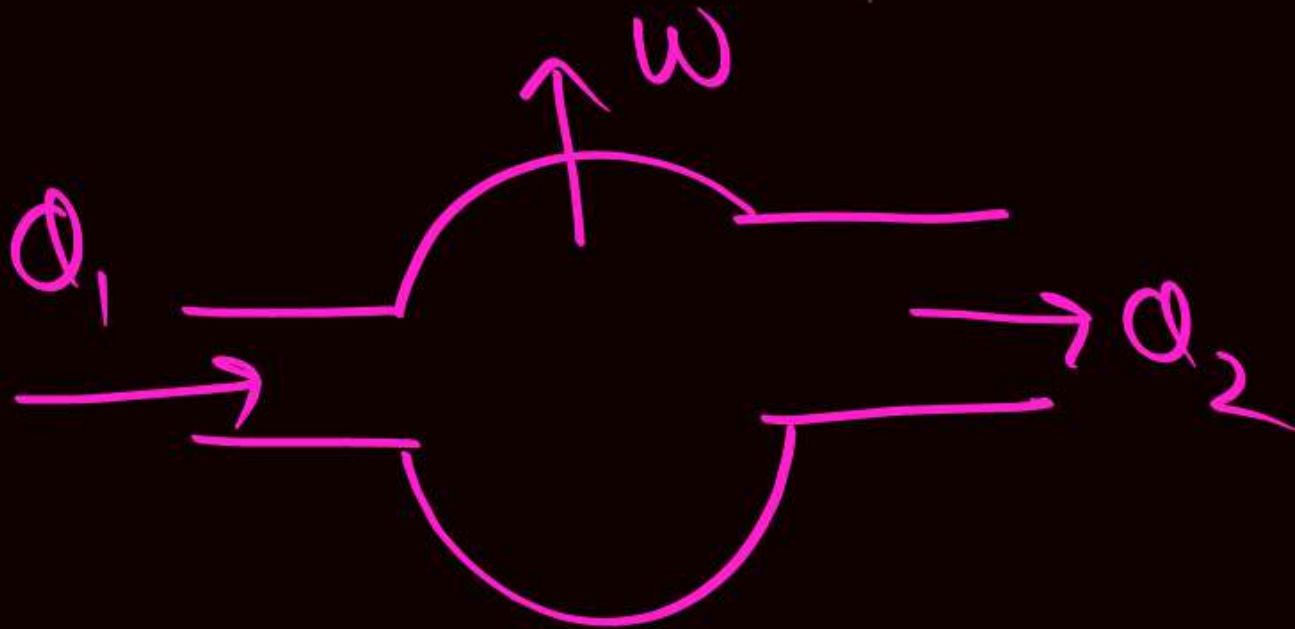
Application of 1st law of thermodynamics

$$\Rightarrow Q = U + W$$

law of conservation
of energy

Application

① Heat engine Q_1



2 mark

Doubt - (3)

(4)

Application of 2nd law of thermodynamics

2 Mark

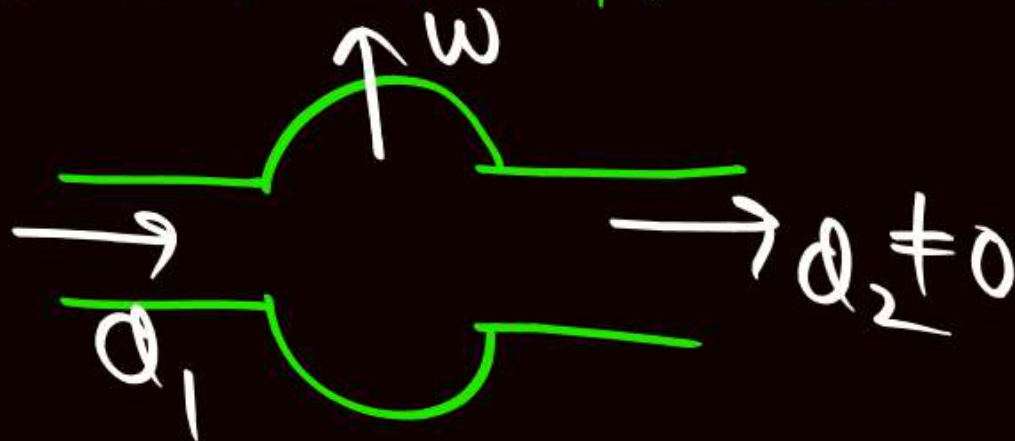
(i) Kelvin and Planck
Statement
or

Heat engine Statement



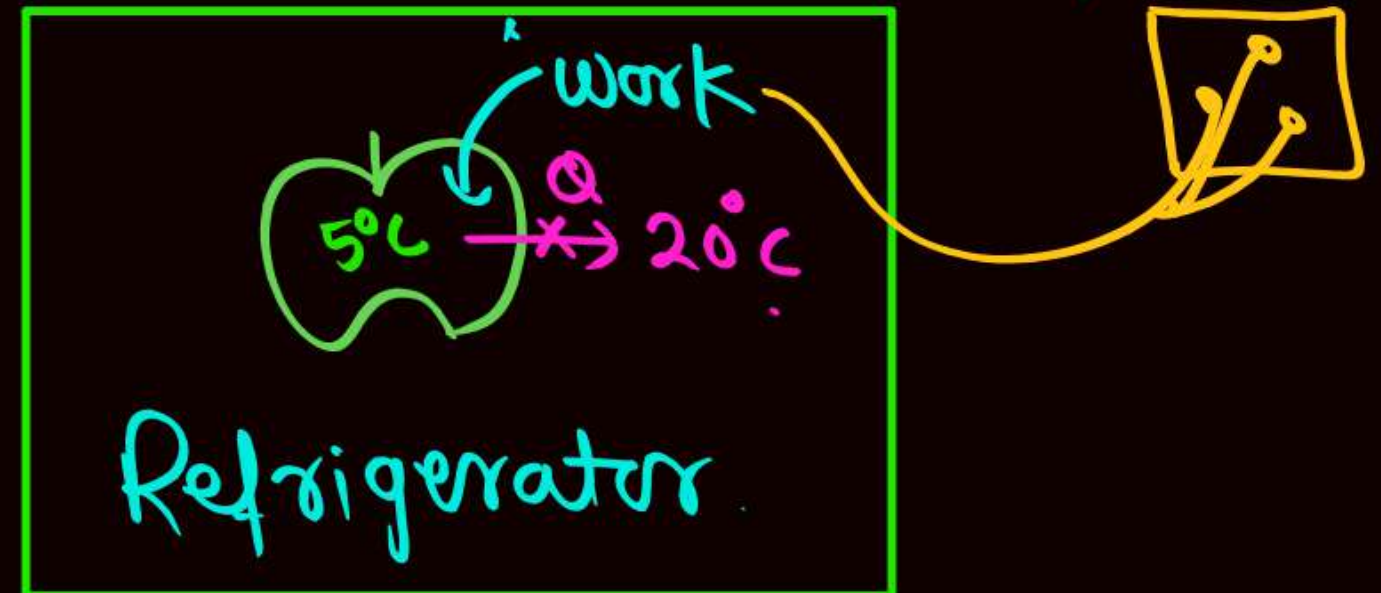
no engine is 100% efficient

Heat engine



(ii) Clausius Statement
or Refrigerator Statement

• No heat travel from Cold Body to hot Body if it do so External work has to be done.



Doubt - (4)

5

Q what is Reversible process? is Reversible process possible? Explain in brief?

1 mark
2 mark

Ans The process which reverse itself



No it is not possible (Because the position of molecule may change)

Gabbbar

Numerical Series

PYQs

Question-01 [JKBOSE Summer 2026 SET-A]



In a cyclic process:

A

The net work done is always positive

B

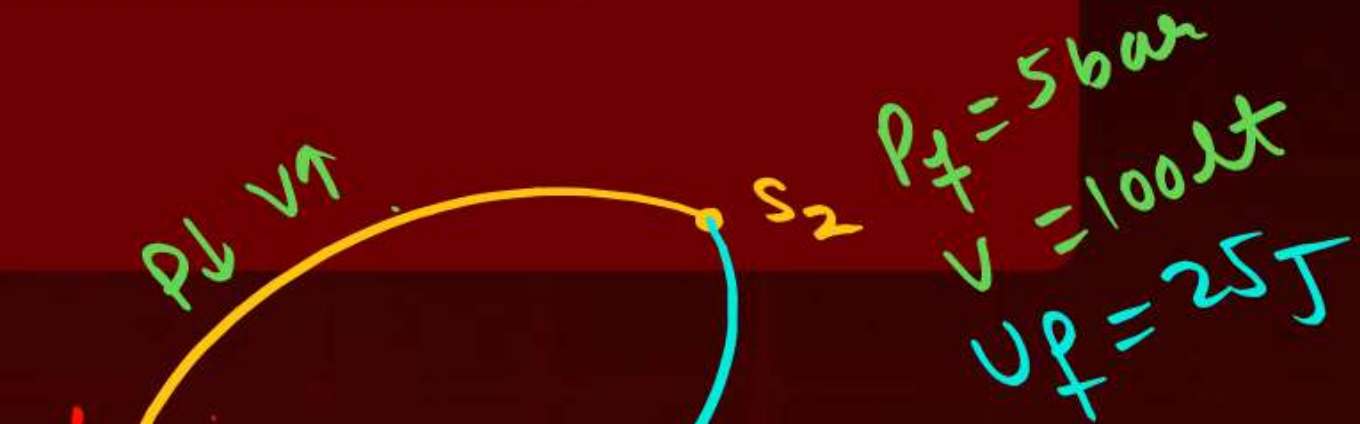
The change in internal energy is zero

C

The net heat transfer is zero

D

The process is always reversible



$$P_i = 10 \text{ bar}$$

$$V_i = 50 \text{ lt}$$

$$U_i = 50 \text{ J}$$

$$U_f = 50 \text{ J}$$

$$\Delta U = U_f - U_i \\ = 50 - 50 = 0$$



Question-02 [JKBOSE Summer 2026 SET-A]

Define isothermal and adiabatic processes and give examples of each. 2M



Question-03 [JKBOSE Summer 2026 SET-A]

State and explain the first law of thermodynamics and discuss its ^{2m}
applications.



Question-04 [JKBOSE Summer 2026 SET-B]

Which of the following variables is a thermodynamic state variable?



Work



Heat



✓ Internal energy



None of these



Question-05 [JKBOSE Winter 2025 SET-A]

The work done in an isothermal expansion of a gas depends upon:



Temperature only



Expansion ratio only



Both of them



None of them

$$W = 2.303 nRT \log \left(\frac{V_2}{V_1} \right)$$

Expansion Ratio.



Question-06 [JKBOSE Winter 2025 SET-A]

Is a reversible process really possible in practice? Discuss briefly.

2M



Question-07 [JKBOSE Winter 2025 SET-B]

The relation of an adiabatic change is:

KT γ Note



$PV = \text{constant}$



$PV^\gamma = \text{constant}$



$PV = RT$



None of them



Question-08 [JKBOSE Winter 2025 SET-B]

What do you mean by reversible process? Give example.



Question-09 [JKBOSE Winter 2025 SET-B]

What are cyclic and non-cyclic processes.



Question-10 [JKBOSE Winter 2025 SET-C]

Repeated



Define Thermal Equilibrium. ✓ How does it occur? ✓ What is the relationship between thermal equilibrium and zeroth law? ✓

2024

2023

2026



Question-11 [JKBOSE Winter 2025 SET-C]

State second law of thermodynamics by Kelvin and Clausius. Give two applications of second law of Thermodynamics.



Question-12 [JKBOSE Winter 2024 SET-A]

State first law of thermodynamics in mathematical form.

$$Q = W + U$$



Question-13 [JKBOSE Winter 2024 SET-B]

When heat is given to a gas in an isobaric process, then:

$$Q_p = U + W$$

A

Work is done by the gas

B

Internal energy of gas increases

C

Both of them

D

None of these



SUMMARY



Doubts
PYQ

HOMEWORK



Copy
check ✓

Mehnat Karo Mehnat karo
Mehnat karne se ho jata hai
Insallah apke 498 No. Aayege
Bas Yhi motivation apne dil me
rkho bs



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You

